



DEPARTMENT OF ECONOMICS
JAHANGIRNAGAR UNIVERSITY
TUTORIAL ANSWER SCRIPT

Class Roll No. 416

Course No. 404

Tutorial No. 02
7261

Course Title: Health economics

Signature of the invigilator:

Date:

ANSWER FROM HERE

Ans. no. 2(a)

Mark Pauly in 1968 gave a model on ~~the~~ how
~~the~~ optimum amount of insurance coverage
is determined in the market. It is known as
Mark Pauly model.

Determination of optimum amount of insurance coverage:-

Let, Elizabeth's total income i.e. her wealth is \$20000. Now if she becomes ill then she will lose \$10,000 when uninsured.

If she buys an insurance with a premium [coinsurance rate 20%] of \$100 and get a coverage of ~~the~~ \$500, then her insurance worksheet, when ill will be :-

Insurance worksheet when ill	Amount
Total income	\$20000
Loss	$\ominus$ \$10000
Remainder	\$10000
Coverage	$\oplus$ \$500
Premium	$\ominus$ \$100
Net income	\$10400

Table:- Insurance worksheet, when ill

Now, the optimum quantity of the insurance is shown in the figure:-

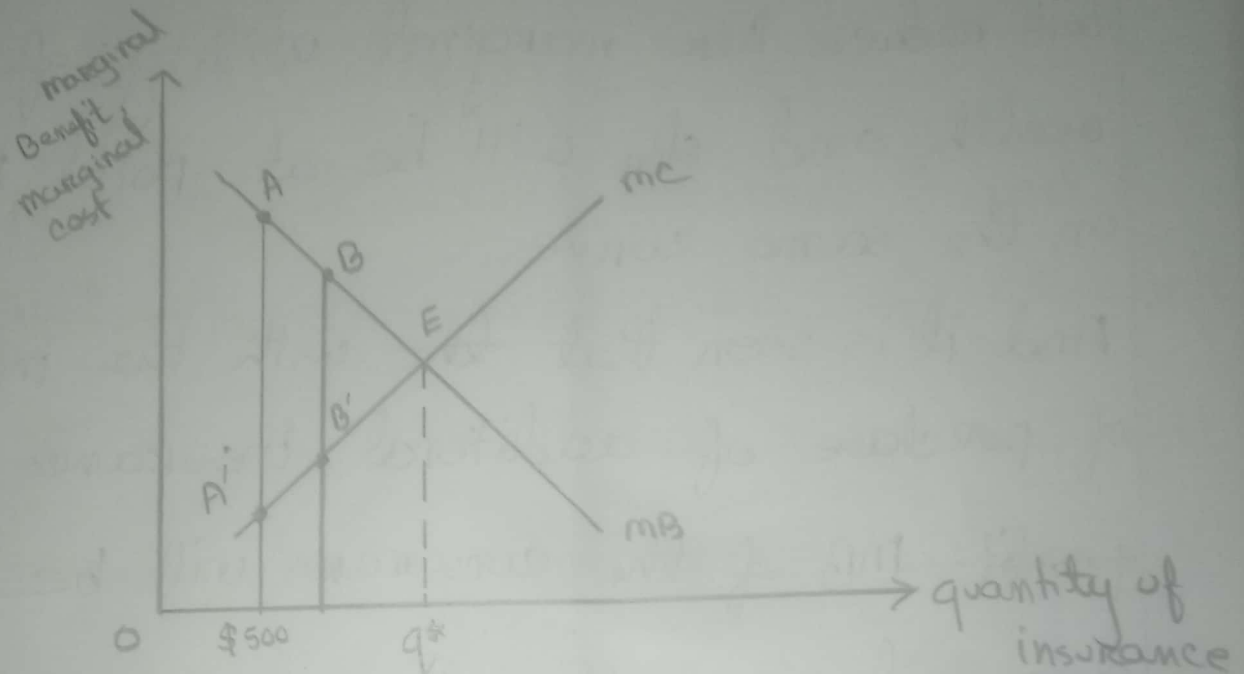


Fig:- optimum amount of insurance

Here, the MB curve indicates the consumer's marginal utility of wealth after buying ~~is~~ getting coverage.

It is seen that Elizabeth when remains healthy, her premium will be \$100, and she will get \$500 coverage and will be

at point A on the MB curve. Now, an additional purchase of insurance will reduce her marginal utility of wealth and she will be at point B on the same curve.

And it is seen that ~~the~~ with the increase of purchase of additional insurance, the ~~quantit~~ MB of the consumers will be decreased gradually. For this reason, the MB curve is downward sloping.

Now, when she gives a premium of \$100, ~~a here~~, MC will be point A'. MC indicates the marginal utility of wealth after buying insurance. When she increases buying insurance then, she

her MC will be increased and new MC will be at B' . Thus the MC curve is upward sloping.

Now, the distance between A and A' indicates the net worth of Elizabeth.

If she is risk averse, then she will increase buying insurance, until the $MB = MC$. At point E, the MB becomes equal to MC. At this point, the quantity of insurance will be q^* .

This is the optimum amount of insurance in the market.

Ans. no. 2(b)

The effect of a change in premiums, expected loss and wealth on insurance coverage is discussed below:-

i) Change in premium:-

Let, there is a coinsurance rate of 20% ~~previo~~ for Elizabeth, and she gets a coverage of \$500. She pays a premium of \$100 and her income is \$20000. If ill she loses \$10000.

Now, the coinsurance rate is increased to 25% for a coverage of \$500.

Now, she has to pay ~~\$10~~ \$125 instead of \$100.

The insurance worksheet is shown below:-

Insurance worksheet when ill	Amount
Income	\$20000
Loss	\$10000
Remainder	\$10000
Coverage	(+) \$500
Premium	(-) \$100 (-) \$125
Net worth	\$10375

Table:- Insurance worksheet, when premium is increased

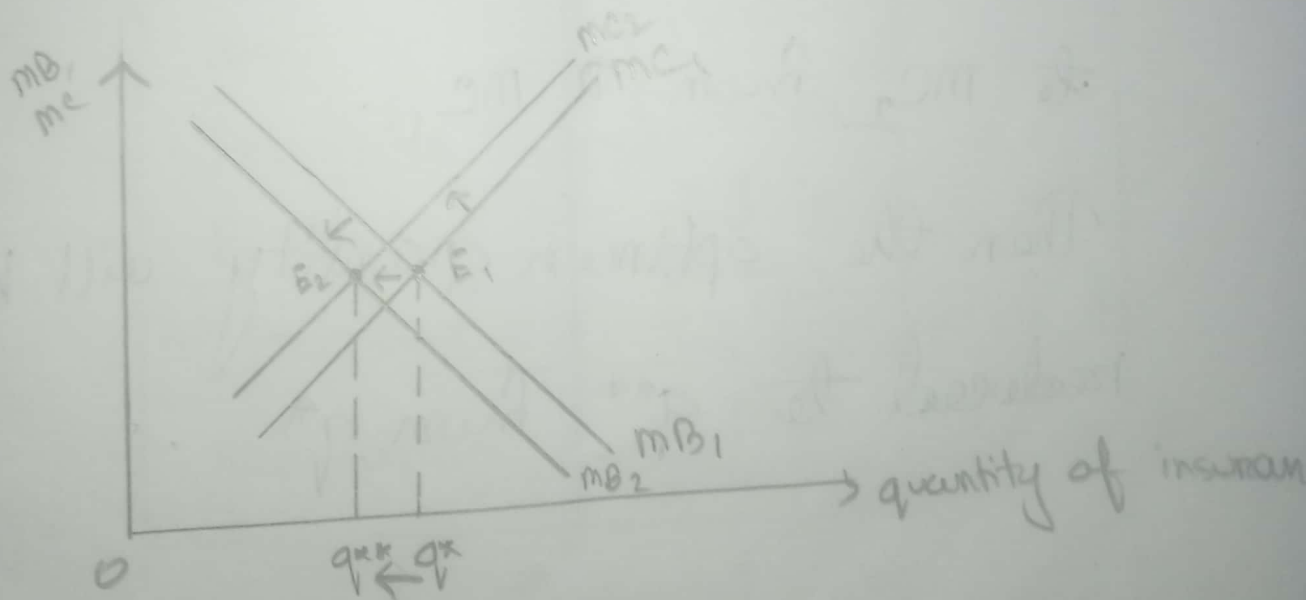


Fig:- change in premium

Here, for the initial premium amount, the ~~MB~~ optimum amount of insurance will be q^* .

Now, when the premium is increased to ~~25%~~ \$125, then the MB of the consumers will be decreased and become MB_2 which is less than MB_1 .

And the MC will be increased to MC_2 from MC_1 .

Then the optimum quantity will be reduced to q^{**} from q^* .

ii) Effect of expected loss:-

Now, the expected loss is increased from \$10000 to \$15000, then the insurance worksheet will ~~be~~ be like following. [holding other things constant]

Insurance worksheet, when ill	Amount.
Income	\$20000
Loss	\$15000
Remainder	\$5000
Pr Coverage	\$500
Premium	\$100
Net worth	\$540.

Fig:- Insurance worksheet, when expected loss is increased

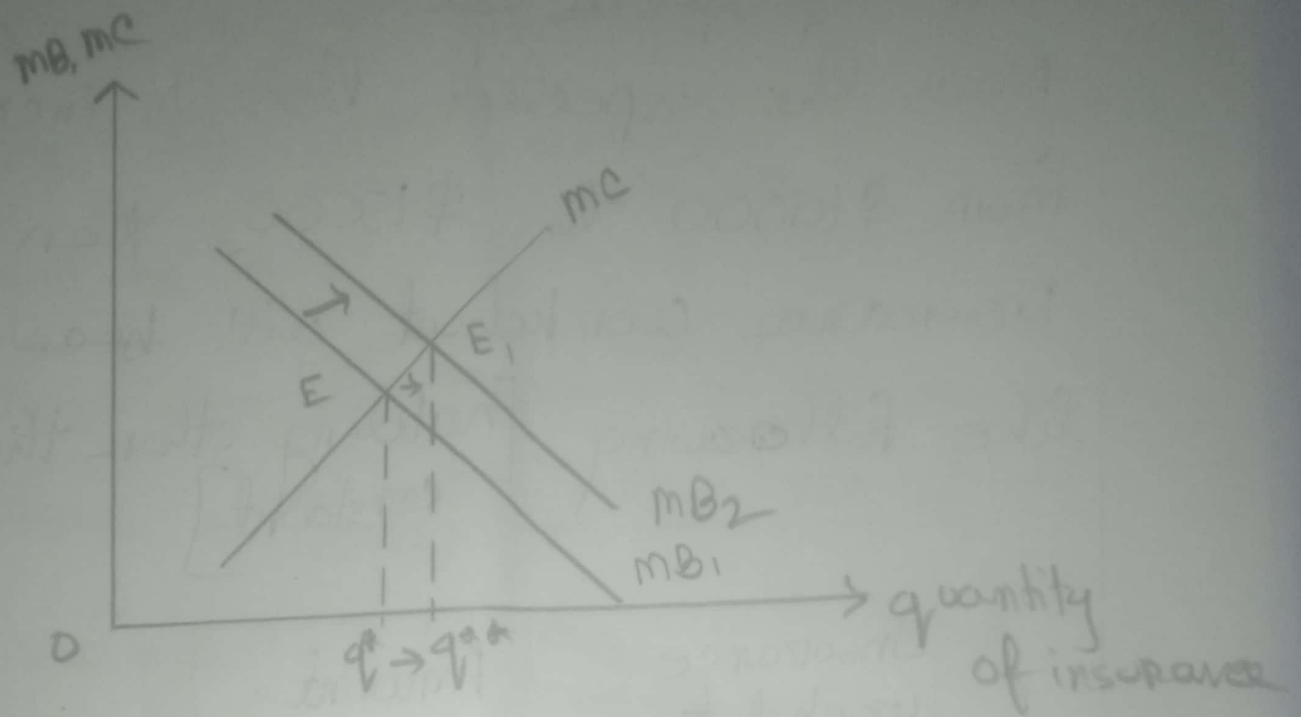


Fig:- change in expected loss

When the ~~of~~ expected loss is increased, then there will be no change in MC.

But the MB will be increased, i.e. the marginal utility of wealth when ill will be increased than before. MB_1 will be shifted to MB_2 and

~~Q~~ q^* will be increased to q^{**} .

New optimum amount of insurance will be q^{**} .

iii) change in wealth:-

Now, suppose the wealth of the consumer is increased to \$25000

Then other things remaining same, here worksheet of insurance will be

Worksheet of insurance, if ill	Amount
Income	\$25000
Loss	\$10000
Remainder	\$15000
Coverage	\$500
Premium	\$100
Net worth	\$15400

Table:- Insurance worksheet, when wealth is increased

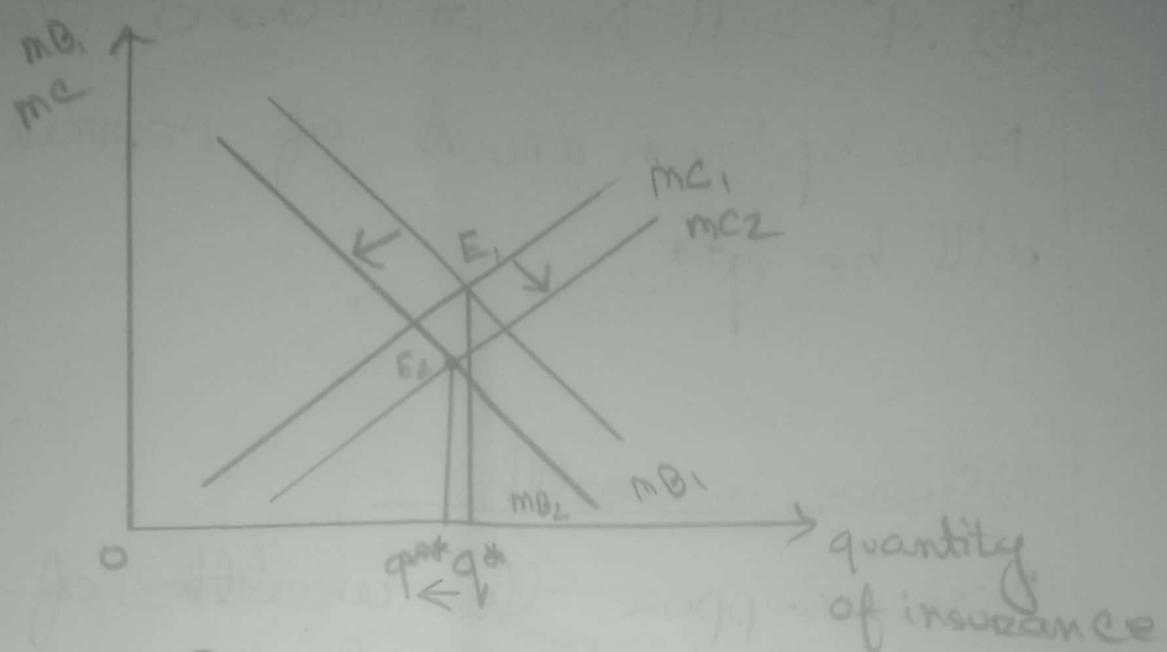


Fig:- Change in insurance when wealth is increased

When wealth is increased, then the MB will be decreased to MB_2 and the MC will be decreased to MC_2 from MC_1 .

Here, the optimum quantity can not be determined exactly whether it will be increased or not, but

In this case the optimum optimum quantity will be slightly less than before. and the optimum amount of insurance will become q^{**} .

These are the effects of change in ~~the~~ premium, ~~the~~ expected loss and wealth on optimum insurance coverage.

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